

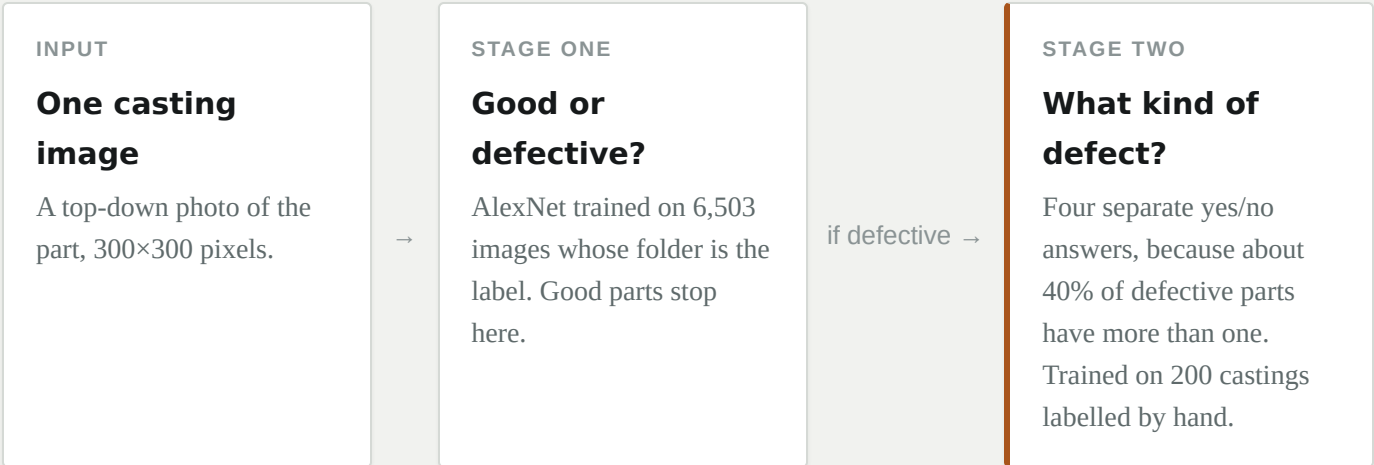
What the inspection model actually learned

The original test set was mostly rotated copies of training images. On a test set built to have none, the good/bad model is still perfect — with or without augmentation — and a second model, trained on 200 hand labels, now says what kind of defect a part has.

7,284 images 716-image leak-free test 200 castings labelled blind four defect types
updated 21 Sept 2026

FINDING 1 89% of the original test set had a rotated twin in training. Hash checks cannot see rotation, so the published split was not really held out.	FINDING 2 On a test set with no twins, 716 of 716 correct. Both models. Those images are about 180–370 different castings, so the true accuracy is at least about 98%.
FINDING 3 Augmentation made no measurable difference. The loss gap in the first version of this report came from two castings.	FINDING 4 A second model names the defect. Four yes/no answers — fray, chip, spot, slash — learned from 200 hand labels, scoring 0.86 to 1.00 AUC.

HOW THE TWO STAGES FIT TOGETHER



FINDING 1

The held-out set was mostly not held out

An earlier check compared SHA-256 hashes and found 64 validation images byte-identical to training images. That check is blind to rotation: turn an image one degree and every byte changes, so a hash reports two different files for what is plainly the same part.

Re-checking with a rotation-proof fingerprint — polar coordinates, then the magnitude of an FFT along the angle axis, which a rotation cannot change — finds far more. Each candidate is then verified by recovering the exact angle and measuring what is left after alignment.

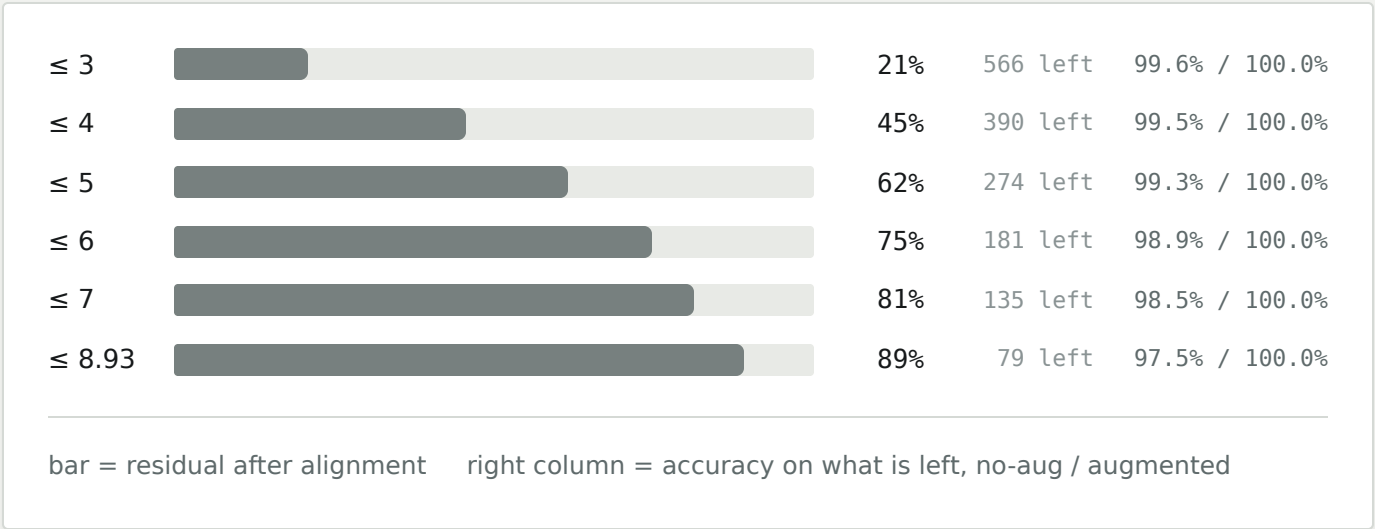
THE NUMBER THAT MATTERS

Splitting the 715 validation castings by whether a rotated copy of them sat in training:

VALIDATION SUBSET	N	NO AUGMENTATION	AUGMENTED
had a rotated twin in training	636	100.00%	100.00%
genuinely unseen	79	97.47%	100.00%
reported overall	715	99.72%	100.00%

The headline 715 of 715 is a blend of an easy subset and a fair one. On castings the model had genuinely never met, the unaugmented model misses 2 of 79 and the augmented one misses none — but with only 79 parts the intervals are wide (± 4 and ± 2 points), so this is not proof that augmentation wins. It is an honest starting point where the 715 figure was not.

HOW THE COUNT MOVES WITH THE THRESHOLD



There is no plateau, so no single number is the answer. Every row sits below where genuinely different castings start (11.2 for defects, 8.8 for good parts), and at a residual of 5 or less the difference images are visually black — impossible for two different physical parts. **At least a third of the dataset is duplicated, plausibly half; and between half and nine-tenths of the validation set has a twin in training.**

11,133 confirmed duplicate pairs	2,092 duplicate families, largest is 7	83.7% of pairs turned more than 10°, so this is rotation, not copying	~3,600 distinct castings behind 7,284 images
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Seventy-nine castings is too thin a test to build on. Finding 2 replaces it with a test set constructed so that no casting has a copy on the other side.

FINDING 2

A test set with no twins, and a perfect score on it

Every image with a confirmed rotated copy was grouped with its copies, and whole groups were assigned to one side of the split. That gives 6,503 training images and 716 test images — 449 defective, 267 good, about the same proportions as the original. A final check counted groups straddling the two sides: zero.

That check trusts the duplicate list, so the split was then tested independently from the pixels. For every test image, the closest training image at any rotation was found and the leftover difference after alignment measured.



bar = share of the test set with a training image this close

right column = accuracy on what is left, no-aug / augmented

Compare the original split, where 89% sat below 8.93. Here 1.3% sit below 3, the level at which two images are visually identical. The higher rows mostly catch *good* castings, which look alike even when they are different parts: in 300 random pairs, two different good castings came as close as 4.7, while two different defective ones came no closer than 10.4. Whatever the cut, the right-hand column does not move.

716 TEST IMAGES**NO AUGMENTATION****AUGMENTED****correct**

716

716

defective parts called good

0 of 449

0 of 449

good parts called defective

0 of 267

0 of 267

716test images, all classified
correctly by both models**177–372**distinct castings behind
those images **$\geq$ 98%**true accuracy, 95%
confidence, counting
castings not images**9**test images visually
identical to one in
training**IMAGES ARE NOT CASTINGS**

Keeping each casting's rotations together is what stops leakage, but it also puts several photos of the same part into the test set side by side. Scoring 716 images is not 716 independent trials. Counting castings instead — somewhere between 177 and 372 depending on how duplicates are grouped — a perfect score still pins the true accuracy above 97.9% at the cautious end and above 99.0% at the other. **“At least about 98%”** is the claim this supports; “at least 99.5%”, which counts images, is not.

THE NINE NEAR-COPIES, AND THE BUG BEHIND MOST OF THEM

Nine test images have a visually identical partner in training. Seven of the nine involve a filename the source dataset uses for two *different* images — it reuses names across its train and test folders — and the split script matched images by filename. That attached some duplicate links to the wrong image, and when two same-named files landed in one folder one overwrote the other: 65 images went missing from training. The script now matches on folder and filename and keeps both copies.

The result does not depend on the nine: both models were also right on all 707 test images that are not near-copies.

WHAT THIS TEST DOES NOT CONTAIN

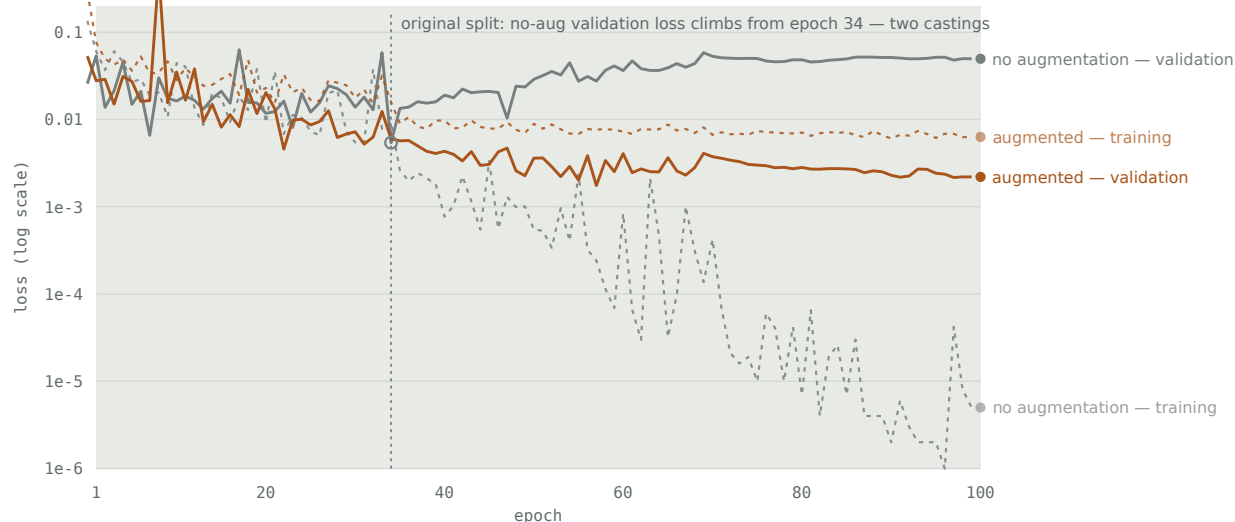
The two castings every unaugmented run missed on the original split (Finding 3) both landed in *training* here, so this test holds no part already known to be hard. The same-casting-two-labels pairs — 7 certain, up to 46 possible, where one photo is filed defective and a rotation of it filed good — also all sit in training. A perfect score on an easy test is still a perfect score, but it cannot show how the model fails, because it did not fail.

FINDING 3

Augmentation made no measurable difference

Two 100-epoch runs, identical except that one augments the training images with flips, rotation and colour jitter. Pretrained AlexNet, batch 16, Adam at $1e-4$. Validation is never augmented. The pair was run on the original split, then again on the leak-free one.

The first version of this report said augmentation stops the model memorising, on the strength of the chart below. The chart is accurate. What it shows turned out to be two castings.



■ no augmentation ■ augmented solid = validation · dashed = training

On the original split, without augmentation, training loss falls to 5×10^{-6} — the model has memorised its training images — and validation loss climbs from epoch 34 onward. That climb looked like general overfitting. It is not. At epoch 100 the total validation loss is 35.5; the 713 correct calls contribute less than 0.04 of it. The rest comes from castings 1591 and 2154, called good at $p = 1.000$ and growing more certain every epoch.

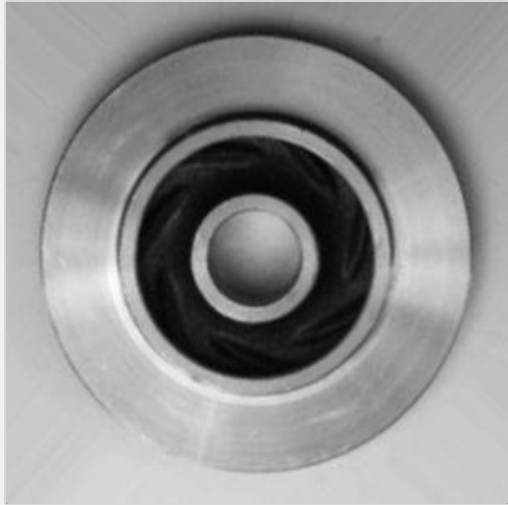
On the leak-free split those two are in training, and the climb is gone:

	NO AUGMENTATION	AUGMENTED
ORIGINAL SPLIT · 715 IMAGES		
accuracy	99.72% (2 missed)	100.00%
final validation loss	0.0496	0.0022
LEAK-FREE SPLIT · 716 IMAGES		
accuracy	100.00%	100.00%
final validation loss	< 0.000001	0.000063
final training loss	< 0.000001	0.0062
least confident call	1.0000	0.956

WHAT CAN BE SAID ABOUT AUGMENTATION

No accuracy benefit was measured on either split. Without augmentation the model memorises its training set and becomes fully certain of every call — its least confident answer on the leak-free test was 1.0000 — and on this data that cost nothing.

The one thing augmentation plausibly buys is doubt. The unaugmented model's two mistakes, below, became impossible to catch with a confidence threshold; the augmented model keeps some uncertainty. That rests on two castings, so it is a reason to leave augmentation on as cheap insurance, not a result.



casting 1591 · called good at $p = 1.000$



casting 2154 · called good at $p = 1.000$

Both are defective; both were called good. The same two failed every unaugmented run. Neither has a rotated twin anywhere in the dataset — they are among the genuinely novel castings, which is why they were hard.

At 15 epochs the model called them wrong at 0.94 and 0.79 confidence, low enough that a review threshold would have caught both. At 100 epochs it called them wrong at $p = 1.000$. Overfitting did not just fail to fix the errors — it destroyed the uncertainty that made them catchable. For an inspection system that is the more expensive failure.

FINDING 4

Defects come in types, and a second model can name them

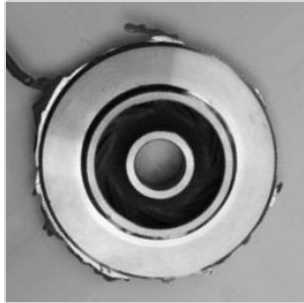
Stripping the final classifier off the trained model exposes a 4,096-number description of each casting. Clustering those splits the 4,211 defective castings in two. To find out what the split means, 200 were labelled by hand with no indication of which group they came from.

Frayed edge

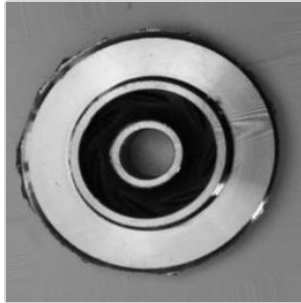
RIM DAMAGE

42.5%

The rim is ragged or torn along its length — a continuous run of disturbed edge, not a single bite out of it.



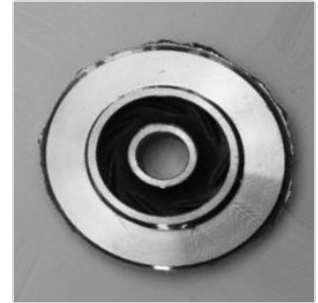
casting 534



casting 7677



casting 9444



casting 1212

Chip

RIM DAMAGE

14.5%

Material missing from the edge: a discrete bite or notch with its own boundary, in one place.



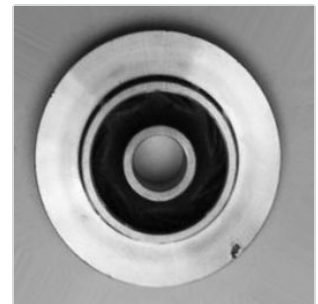
casting 6210



casting 3308



casting 6548



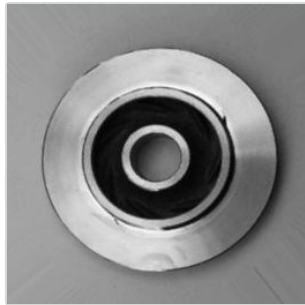
casting 2558

Slash

RIM DAMAGE

35.0%

A linear mark — scratch, gouge or drag line, longer than it is wide.



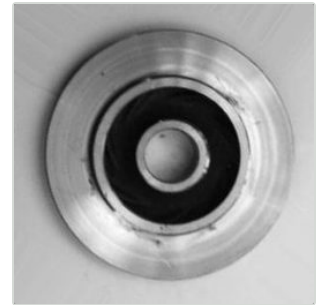
casting 1462



casting 5251



casting 2158



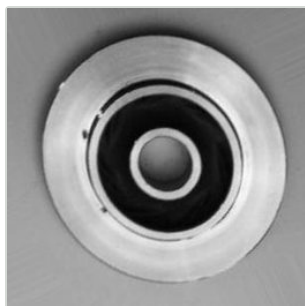
casting 552

Spot on the face

FACE DAMAGE

48.5%

A blemish on the flat face away from the rim — porosity, an inclusion, a blotch. The rim itself is usually clean.



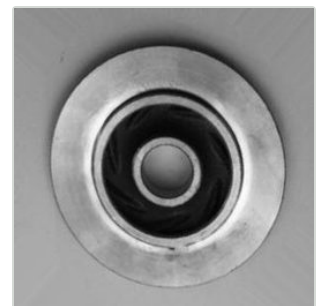
casting 387



casting 7298

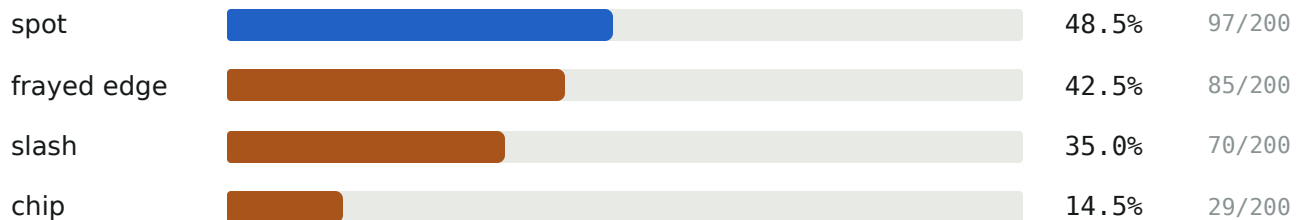


casting 9604



casting 9808

HOW COMMON EACH ONE IS

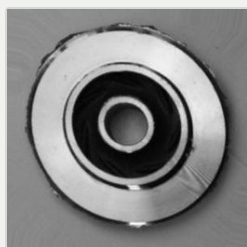


rim damage face damage Bars exceed 100% because castings carry more than one.

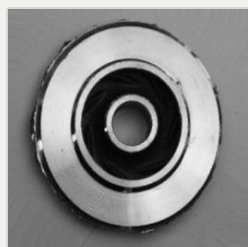
MOST DAMAGED PARTS HAVE SEVERAL THINGS WRONG

15 show no visible defect at all	111 have exactly one defect type	74 have two or more — 37% of the sample	30 have rim damage and a face spot together
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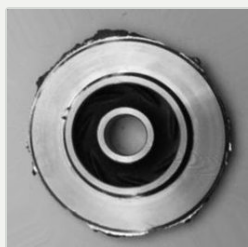
This is why the right output is four yes/no questions, not one four-way choice. A model forced to pick a single label is wrong by construction on more than a third of the parts.



casting 443



casting 8482



casting 2924



casting 9929

WHAT THE GROUPING ACTUALLY PREDICTS

DEFECT	GROUP A 1,328 parts	GROUP B 2,883 parts	GAP	P
frayed edge	58%	27%	+31	<0.0001
slash	45%	25%	+20	0.005
chip	20%	9%	+11	0.043
spot on face	40%	57%	-17	0.023

Three rows lean one way and the fourth leans the other. That opposition is what makes this a distinction between *types*. A model merely ranking parts from lightly to badly damaged would push every row the same direction. Collapsed to families: any rim damage — A 77%, B 41% against face spot — A 40%, B 57%.

RULING OUT “JUST MORE DAMAGED”

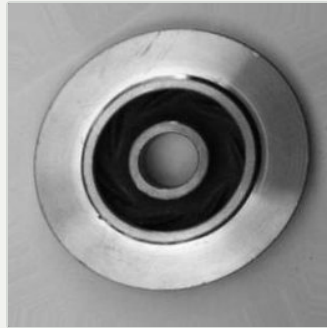
Take only the 111 castings with exactly one defect. Every part in that set carries the same amount of damage, so severity is held constant by construction and anything left is about type.

Group A is **45% face spots**, group B is **73%** — $p = 0.003$. The split survives with severity fixed.

FIFTEEN CASTINGS WITH NOTHING WRONG



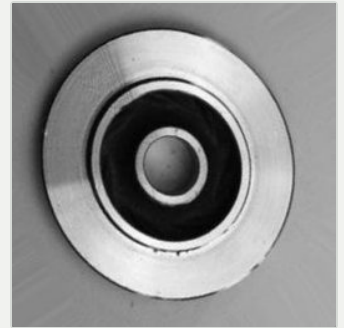
casting 732



casting 4642



casting 8111



casting 9785

Every image in this study is labelled defective in the published dataset. On 15 of 200 — about **10% of all defective castings** once scaled — no defect was visible. Either those labels carry noise or the defect is too subtle to see at this resolution. Both belong next to any accuracy figure: a classifier cannot be separating defective from good more cleanly than the labels themselves distinguish them.

THE FIRST MEASUREMENT GOT THIS BACKWARDS

An earlier round asked one question covering “*spots, chips or nicks*” together. It was true of 89% of castings — far too common to separate anything — and it pointed the opposite way to its own spot component, because chips and slashes lean toward group A and outvoted spots, which lean toward B.

As asked: A 96% / B 81%. The spot part alone, once separated: A 40% / B 57%. A combined variable inverted a real effect, and the conclusion drawn from it — that the model tracked severity — was wrong.

That round was not wasted. Its frayed-edge question used the same narrow wording both times and replicates: +22.5 points then, +31.0 now, pooling to +28.3, $p \approx 0.000001$ over 298 castings. Seven castings were drawn in both rounds and labelled twice, blind, weeks apart — called the same way 7 of 7.

TEACHING A SECOND MODEL THE FOUR TYPES

The 200 hand labels became training data for stage two: the same network shape, but ending in four independent yes/no outputs instead of one choice. It was scored by five-fold cross-validation, with every rotation of a casting kept in the same fold so it is never tested on a copy of something it trained on.

DEFECT	IN THE 200	AUC	HOW FAR TO TRUST IT
frayed edge	85	1.00	reliable
spot on face	97	0.92	good
slash	70	0.92	good
chip	29	0.86	weakest — fewest examples

AUC is the chance that a randomly chosen casting *with* the defect scores higher than one without it: 0.5 is guessing, 1.0 is perfect.

CHECKING THE PERFECT SCORE

A fray AUC of 1.00 looked too good, so the fray labels were shuffled among the 200 castings and the whole cross-validation run again. If anything in the pipeline leaked the answer, the shuffled run would still score well. It scored **0.46** — chance. The model learned frayed edges from the images.

Two other checks came back negative. Going from 140 to 200 labels improved AUC by less than 0.01, so another labelling round would add little. Starting stage two from the good/bad model's weights instead of the ImageNet ones was worse on three flags of four.

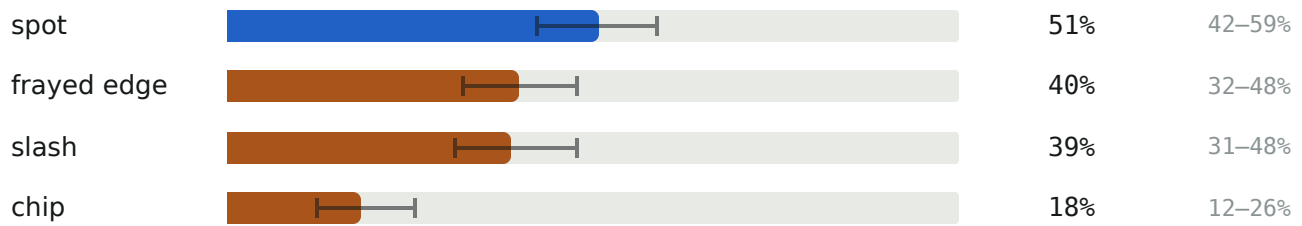
APPLIED TO ALL 4,211 DEFECTIVE IMAGES

DEFECT	HAND LABELS, REWEIGHTED	MODEL, ALL 4,211	MODEL LEANS
spot on face	51.6%	52.7%	+1.1
frayed edge	36.8%	34.0%	-2.8
slash	31.3%	36.3%	+5.0
chip	12.5%	20.3%	+7.8

The hand-label column is reweighted because the 200 were drawn 100 from each cluster while the clusters hold 1,328 and 2,883 images. Against it, the model calls chips and slashes too often. That is the side-effect of the training setting that stops rare defects being ignored, and it is why chip and slash figures below are quoted with the lean taken off.

ON DEFECTIVE CASTINGS THE MODEL HAS NEVER SEEN

The leak-free test holds 449 defective images. Dropping 9 with a training image closer than 5.7, and 6 whose filename is shared with a different image, leaves 434 — between 126 and 223 distinct castings. The ranges below are computed at 126, the cautious end.



bar = share the model calls present whisker = 95% range, counting castings

with the lean removed: spot ≈50%, fray ≈43%, slash ≈34%, chip ≈11%

55%

have exactly one defect
type

32%

have two

9%

have three or all four

5%

have none of the four,
though filed defective

The commonest pattern is a face spot on its own, in 41% of these images. Next is frayed edge *and* slash together, in 29% — the two travel as a pair. That is not the model confusing them: in the hand labels, 57 of the 85 frayed castings also had a slash. The 5% with nothing found echoes the 15 of 200 hand-labelled castings that showed no visible defect.

Every rate here is within its range of the figure across all 4,211 defective images, so the unseen set looks like the rest of the data rather than an unusual corner of it.

WHAT THIS SUPPORTS, AND WHAT IT DOES NOT

SUPPORTED

The published split is contaminated by rotated duplicates that a hash check cannot detect. On a test set with none, the good/bad model made no errors on 716 images from roughly 180–370 castings: true accuracy at least about 98%. Defects divide into rim damage and face spots, which the model found unsupervised and a blind human sort reproduces. Stage two identifies frayed edges reliably, spots and slashes well, and chips less well.

NOT SUPPORTED

That augmentation improves accuracy — no difference on either split. Any accuracy figure on the original split. Exact defect-type percentages: they are model estimates with a known lean, chip most of all. Anything that needs a second opinion on the labels: all 200 come from one labeller, consistent with himself (7 of 7 on repeats) but not checked against anyone else.

NEXT

Join both stages into one prediction call that returns, per casting, good or defective plus the four defect answers with their confidence — the record a web front end would consume. Have a second person label 50 of the 200 to measure agreement; it is the cheapest way to strengthen stage two. Optionally rebuild the split with the fixed script and retrain, to remove the nine near-copies and restore the 65 lost training images; the result above does not depend on either.

Stage one: pretrained AlexNet, 100 epochs, batch 16, Adam $1e-4$, with and without augmentation, on the original split and on a group-aware split (6,503 / 716). Duplicates by rotation-invariant polar-FFT fingerprint with phase-correlation alignment; the split was re-checked against every training image at its recovered rotation. Stage two: AlexNet with four sigmoid outputs, 40 epochs, five-fold group-aware cross-validation on 200 castings labelled blind across two rounds, with a label-shuffling control. Casting counts use union-find (floor) and complete-linkage (ceiling) over confirmed duplicate pairs; intervals are Wilson at 95%.